

Design & Analysis of Algorithms

Lab 1: Algorithm Performance Analysis Workbench

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Q1. 1.1 Recursive and Iterative Computation Analyzer (10 marks)

```
def analyze_recursive_iterative(n):
    def recursive_factorial(x):
        if x==0:
            return 1
        return x*recursive_factorial(x-1)

    def iterative_factorial(x):
        result = 1
        for i in range(1,x+1):
            result*=i
        return result

    fib_count = 0

    def recursive_fibonacci(x):
        nonlocal fib_count
        fib_count+=1

        if x==0:
            return 0
        if x==1:
            return 1
        return recursive_fibonacci(x-1)+recursive_fibonacci(x-2)

    def iterative_fibonacci(x):
        if x==0:
            return 0

        a=0
        b=1
        for i in range(x):
            a,b=b,a+b
        return a

    rf = recursive_factorial(n)
    itf = iterative_factorial(n)
    rfib = recursive_fibonacci(n)
    ifib = iterative_fibonacci(n)

    return["Computation Analysis Report",
        f"Recursive Factorial: {rf}",
        f"Iterative Factorial: {itf}",
        f"Recursive Fibonacci: {rfib}",
        f"Iterative Fibonacci: {ifib}",
        "Operation Count Comparison",
        f"Recursive Factorial Count: {n+1}",
        f"Iterative Factorial Count: {n}",
        f"Recursive Fibonacci Count: {fib_count}",
        f"Iterative Fibonacci Count: {n}"
    ]
```